Subject outline

Lecture topics (by week)

1 – Introduction to quantum computing and maths basics
2 – Single qubit representations and logic operations
3 – Two qubit states and logic gates
4 – Multi-qubit states and quantum arithmetic
5 – Basic quantum algorithms
6 – Period finding, cryptography and quantum factoring
7 – Shor's algorithm, post-quantum crypto, quantum key distribution
8 – Quantum search algorithms
9 – Grover search applications, optimisation problems
10 – Solving optimisation problems on quantum computers
11 – Applications in quantum machine learning
12 – Real quantum computer devices

Assignment schedule:
#1: Hand out in Week 2
#2: Hand out in Week 8

Week 8

Lecture 15

15.1 Grover's algorithm: Overview and intuition

15.2 The Oracle

15.3 Inversion about the mean

15.4 Example Quantum Search

Lecture 16

16.1 Grover's algorithm Example

16.2 Probability of success

16.3 Geometric interpretation

Practice class 8

Grover's algorithm using QUI

Recap: Post-Quantum Cryptography & QKD

Post-Quantum Cryptography

Table 1 - Impact of Quantum Computing on Common Cryptographic Algorithms

Cryptographic Algorithm	Type	Purpose	Impact from large-scale quantum computer
AES	Symmetric key	Encryption	Larger key sizes needed
SHA-2, SHA-3	----------	Hash functions	Larger output needed
RSA	Public key	Signatures, key establishment	No longer secure
ECDSA, ECDH (Elliptic Curve Cryptography)	Public key	Signatures, key exchange	No longer secure
DSA (Finite Field Cryptography)	Public key	Signatures, key exchange	No longer secure

Quantum Key Distribution

Alice Sends	Alice Basis	Alice Sends	Eve's Basis	Eve measures	Eve sends	Bob's Basis	Bob measures	A/B Same basis?	Do their bits match
0	x	|+>	x	0	|+>	x	0	✓	✓
1	z	|1>	z	1	|1>	x	0	✗	-
1	x	|->	z	1	|1>	z	1	✗	-
0	z	|0>	x	0	|+>	z	1	✓	✗

BB84 Protocol

15.1 Grover's algorithm: Overview and intuition

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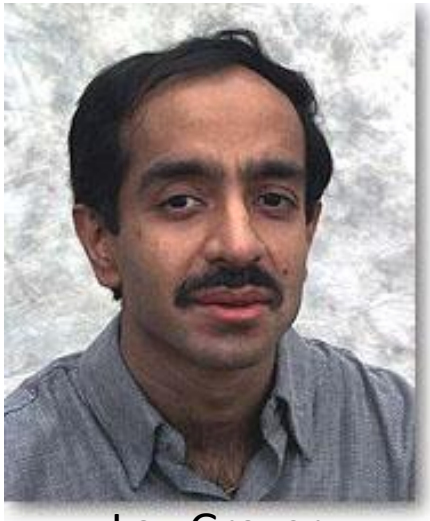
Lecture 15

Grover's Algorithm (1996)

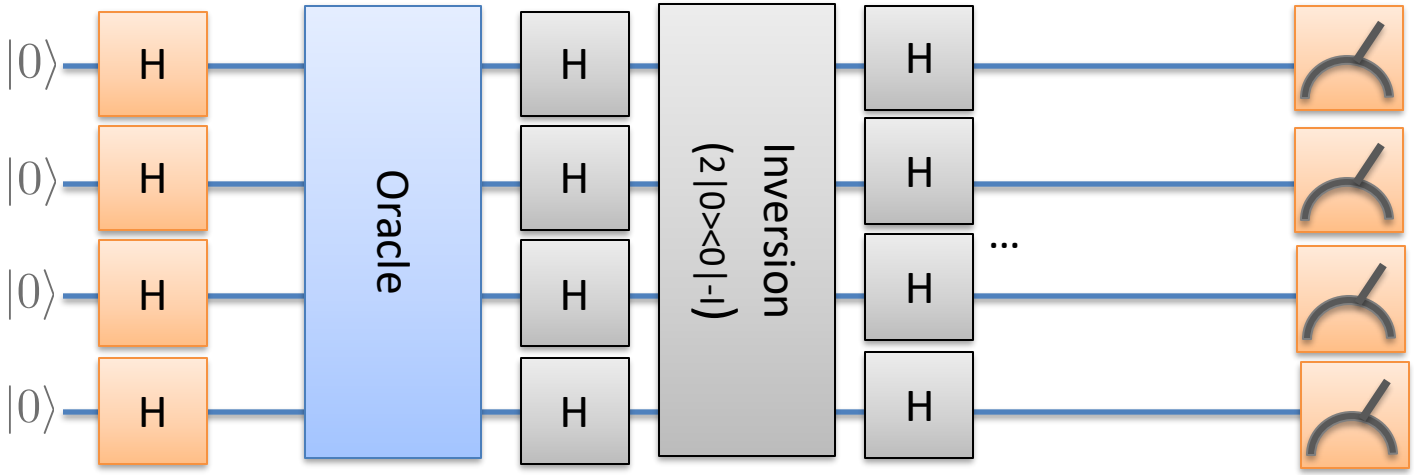
- Unordered search, find marked item among many
- Classically, this requires N/2 queries to the oracle
- Quantum mechanically, requires only O(√N) queries.



Lov Grover

Simple problem = search for one integer marked by the oracle.

High level structure:



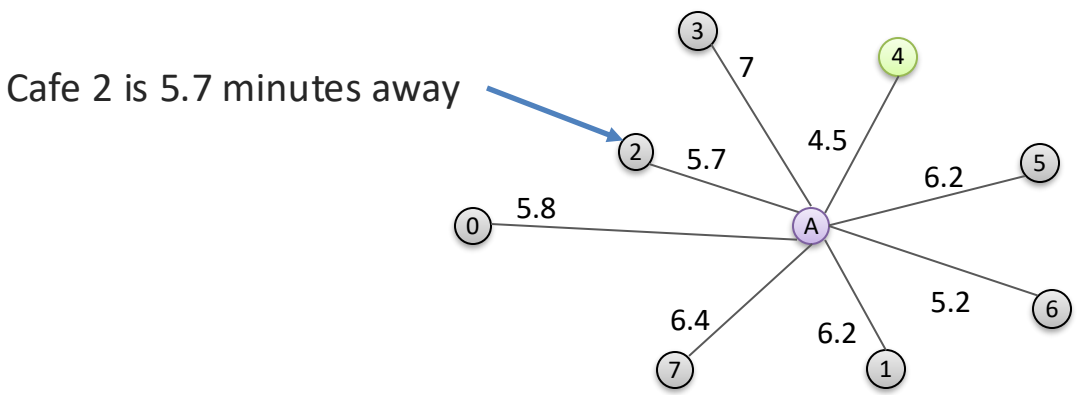
Unstructured search problems

Given a set of N elements $X = \{x_0, x_1, \dots, x_{N-1}\}$
and a Boolean function $f: X \to \{0,1\}$,
find an element $x \in X$ such that $f(x) = 1$.

- Unstructured means that we cannot assume how $x_0, x_1, \dots, x_{N-1}$ are ordered.
- The function f verifies the property that we are interested to find.
  - Important: The function f *does not compute a solution*, it only verifies if the given input satisfies the property of interest.
  - The function f can be thought as an "oracle" that verifies if an input is a solution.

Unstructured search problems

- Example: Alice (A) wants to find a coffee shop within 5 mins of walking distance.

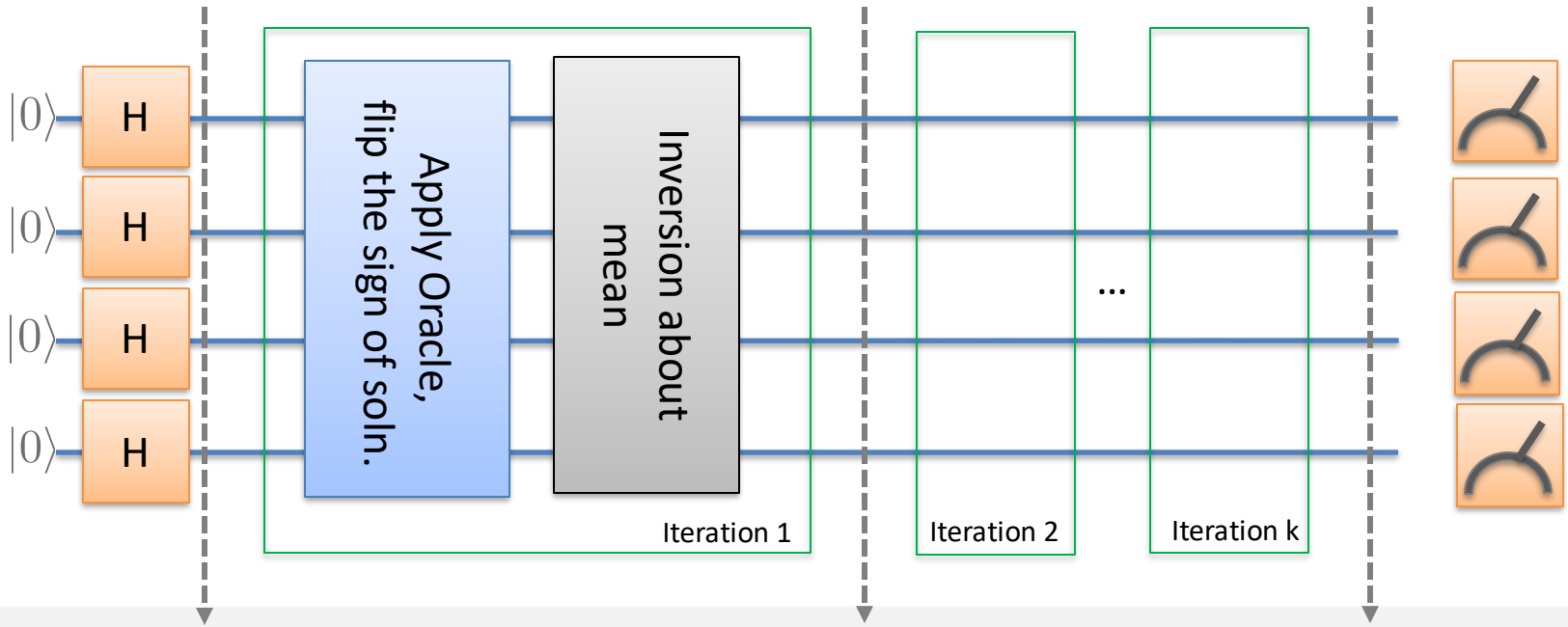


- Set $X = \{0,1,2,\ldots,7\}$ denotes the search space
  - Set X is not ordered by travel time
- $f(x) = \begin{cases} 1, \text{ if time(x)} \leq 5 \\ 0, \text{otherwise} \end{cases}$
  - In general, the function f does not need to know about the search space.
- Solution x=4 (Cafe 4 is the only one that is less than 5 minutes away)
- The function f is like a blackbox (similar to Deutsch-Jozsa and Simon's algorithms).

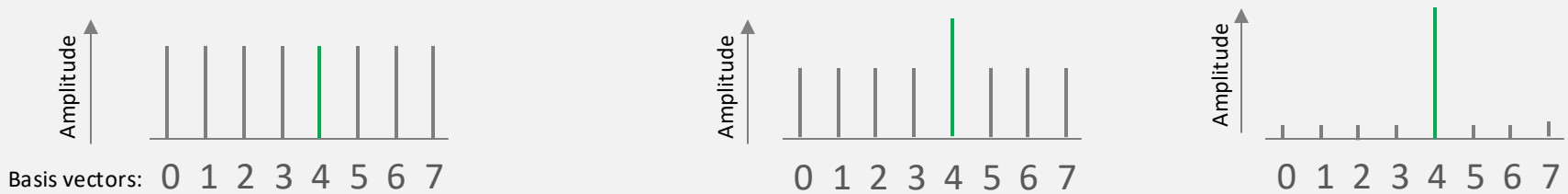
Grover's algorithm: Intuition

Equal superposition

Solution has higher probability

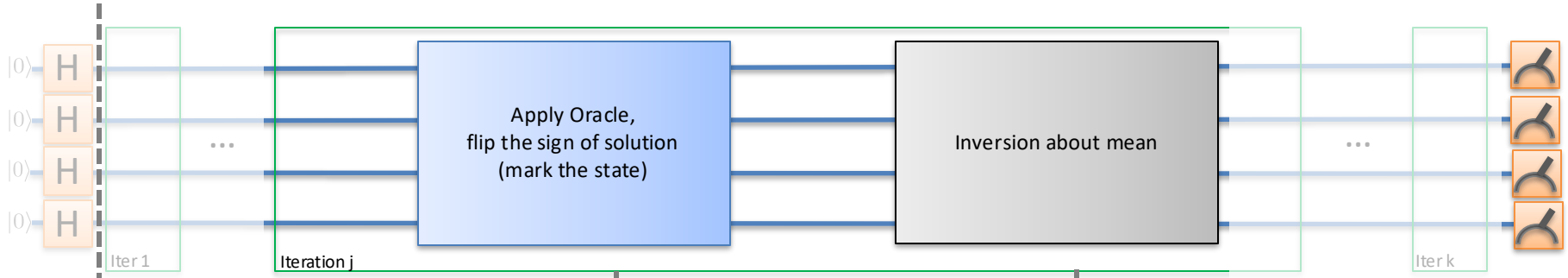


Alice's coffee problem



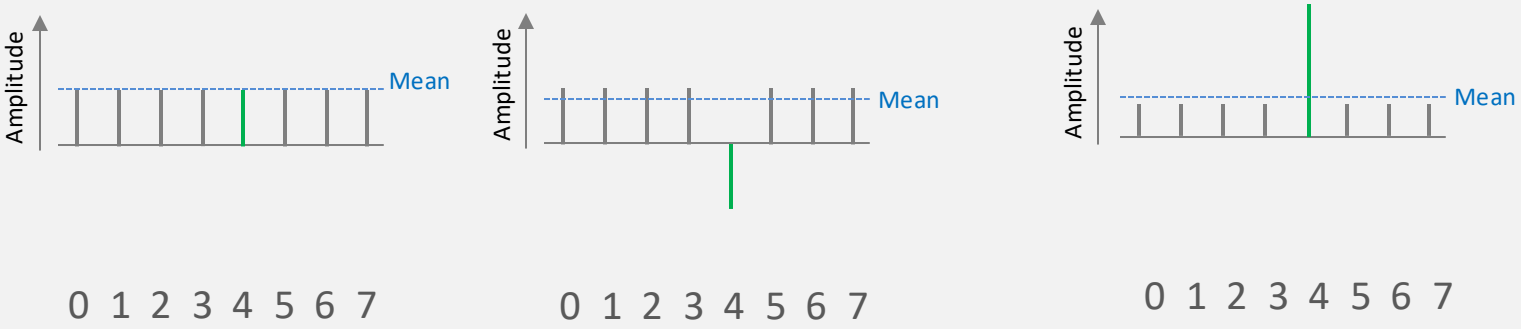
Intuition: Each iteration amplifies the amplitude of solution vectors.

Grover's algorithm: Intuition



Maps every amplitude Mean$-\delta$ to Mean$+\delta$

Alice's coffee problem



Intuition:

Flipping the sign of the amplitude of solution vector reduces the mean.

Inversion about the mean increases the amplitude of solution vector.

15.2 The Oracle

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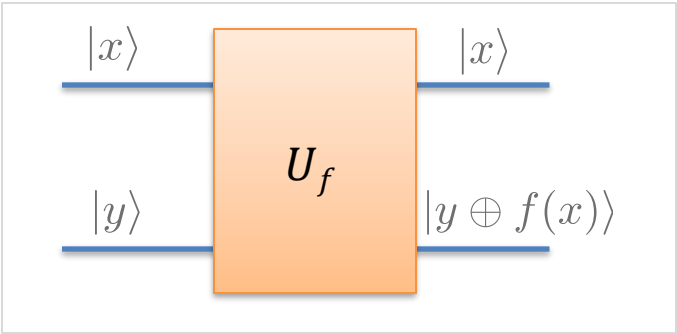
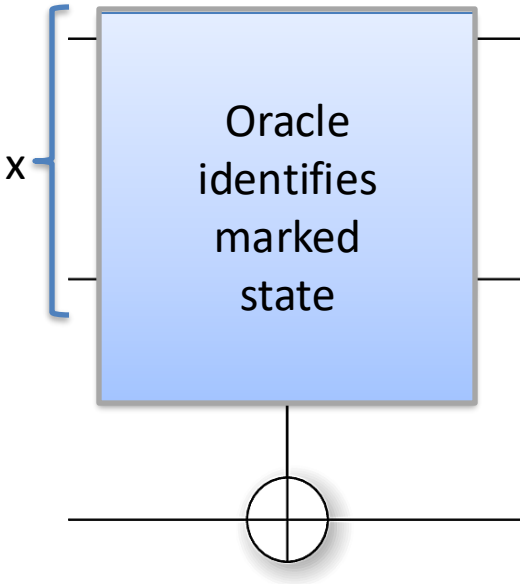
Lecture 15

The Oracle

The task of recognizing the correct solution goes to the “oracle”.



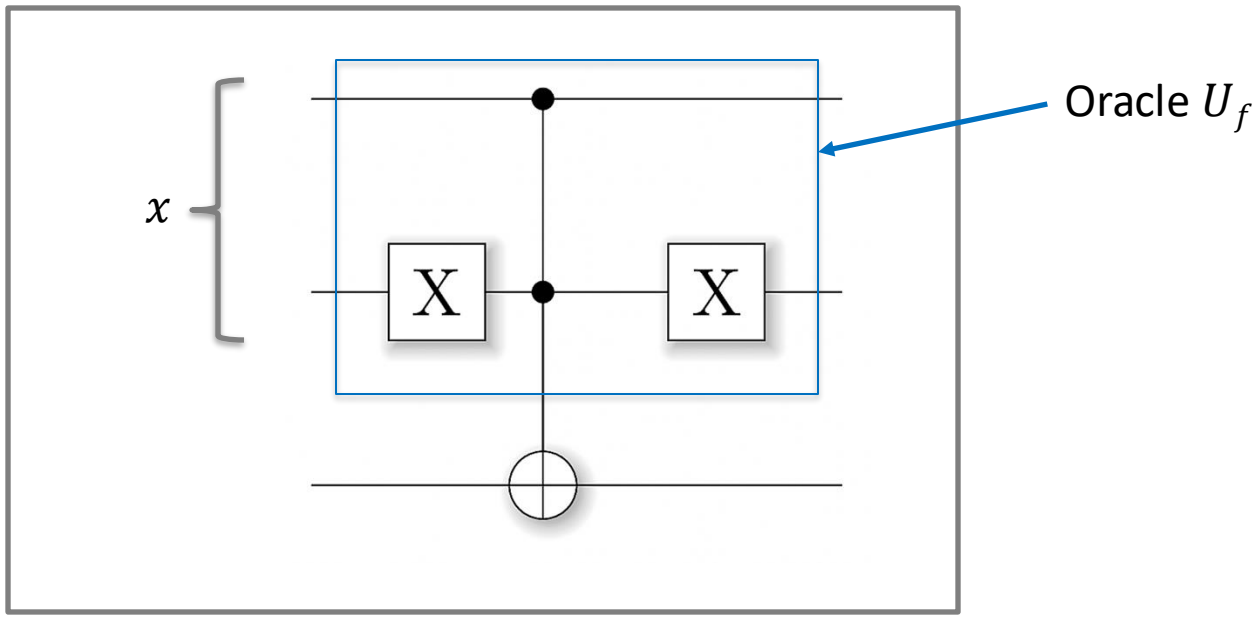
Binary function, or “oracle”
Identifying a marked state, m



Designed to flip the last bit if the input, x, is a solution

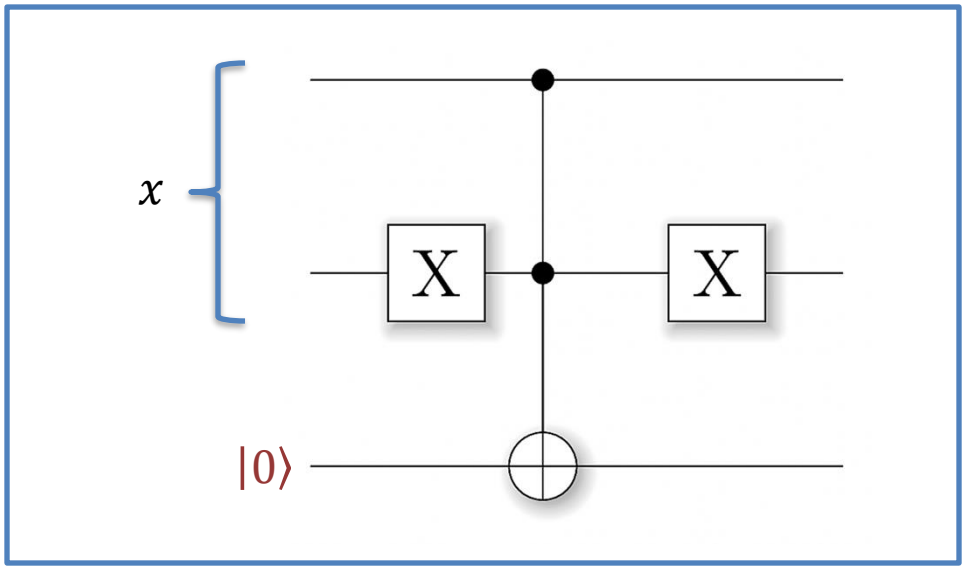
The oracle is just a Boolean function (as seen in previous lectures)

An example Oracle



The oracle U_f flips the target qubit if the input, x = 10 (i.e., 2 in decimal).

Oracle on $|0\rangle$ state



Let's see what happens when we apply to an equal superposition (in x register):

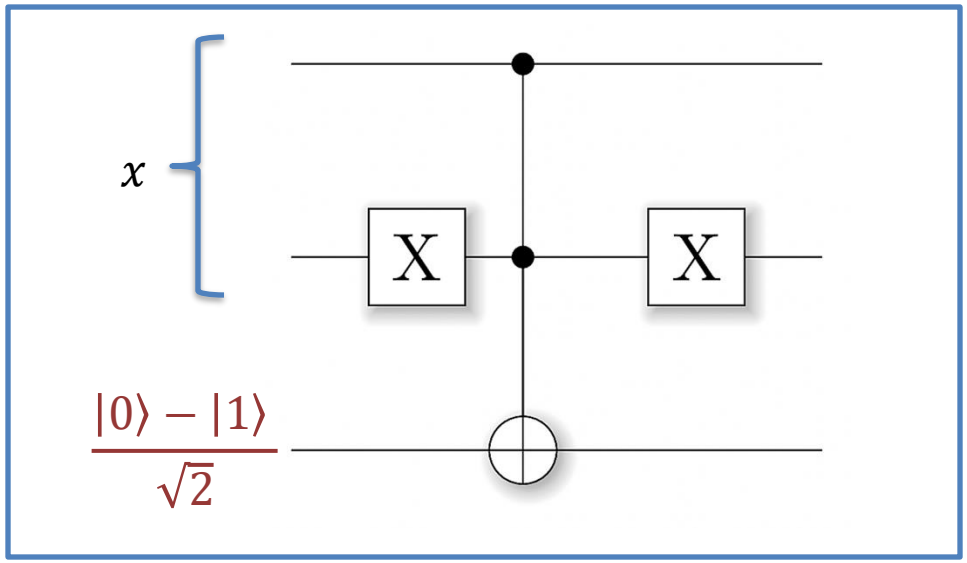
$$|x\rangle|0\rangle = \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2}|0\rangle$$

$$= \frac{|00\rangle|0\rangle + |01\rangle|0\rangle + |10\rangle|0\rangle + |11\rangle|0\rangle}{2}$$

$$U_f(|x\rangle|0\rangle) = \frac{|00\rangle|0\rangle + |01\rangle|0\rangle + |10\rangle|1\rangle + |11\rangle|0\rangle}{2}$$

Identifies the solution state, but does not flip its sign (yet).

Oracle on equal superposition state



Acting X gate on target qubit: $\mathrm{X}|-\rangle = \mathrm{X}\frac{|0\rangle - |1\rangle}{\sqrt{2}} = \frac{|1\rangle - |0\rangle}{\sqrt{2}} = -|-\rangle$

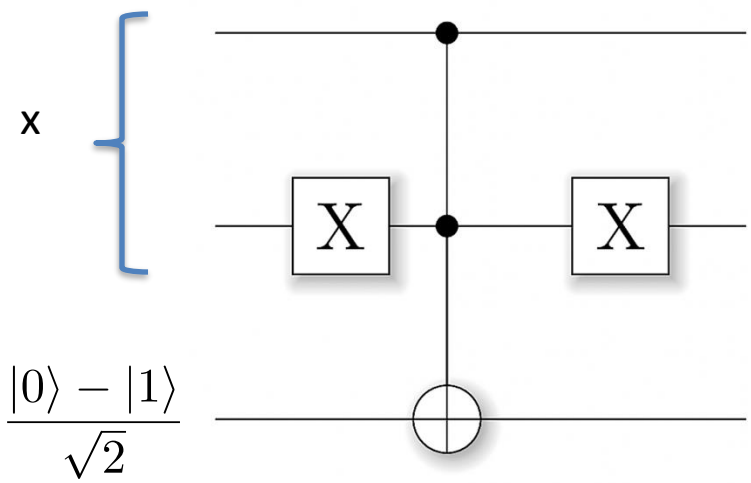
$$|x\rangle|-\rangle = \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2}|-\rangle$$

$$U_f(|x\rangle|-\rangle) = \frac{|00\rangle|-\rangle + |01\rangle|-\rangle - |10\rangle|-\rangle + |11\rangle|-\rangle}{2}$$

Marks the solution state, and flips its sign.

Oracle qubit unchanged!

Example: Oracle recognizing the state "2 = $|10\rangle$"



<u>Phase kickback</u>

$$|00\rangle \rightarrow \quad |00\rangle$$
$$|01\rangle \rightarrow \quad |01\rangle$$
$$|10\rangle \rightarrow -|10\rangle$$
$$|11\rangle \rightarrow \quad |11\rangle$$

The effect on each of the 4 states in the 2-qubit control register, x:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The marked state

In general: $|x\rangle|-\rangle \xrightarrow{U_f} (-1)^{f(x)}|x\rangle|-\rangle$

- If $f(x) = 0$, $|x\rangle|-\rangle \xrightarrow{U_f} |x\rangle|-\rangle$
- If $f(x) = 1$, $|x\rangle|-\rangle \xrightarrow{U_f} -|x\rangle|-\rangle$

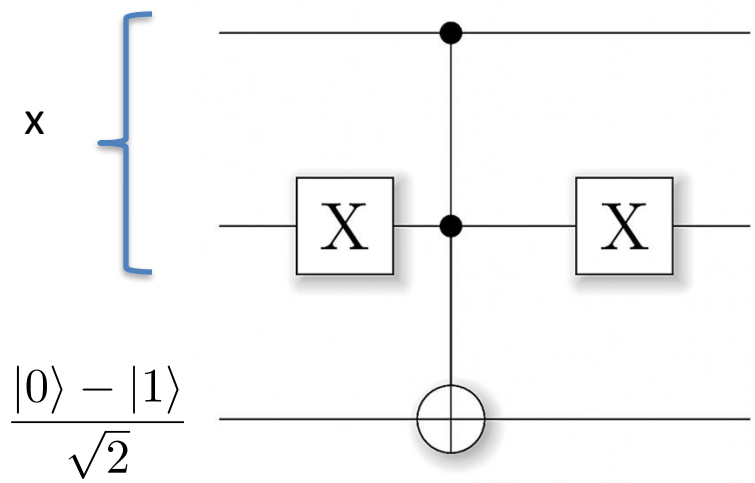
Initially in Grover's algorithm, we will be searching for a *single (integer) solution*, ***m***. In that case the effect of the oracle on the control register is:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

-1 in the mth position

Here, as in future slides, we are only writing out the control qubits (in this case 2 qubits only).

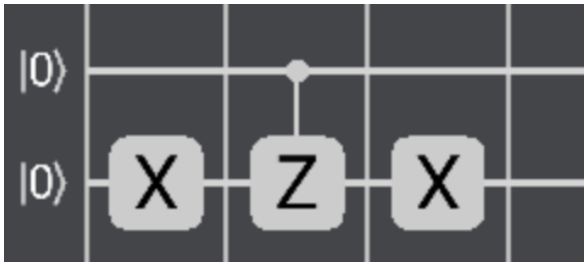
Example: Oracle recognizing the state "2 = $|10\rangle$"



<u>Phase kickback</u>

$$|00\rangle \rightarrow \quad |00\rangle$$
$$|01\rangle \rightarrow \quad |01\rangle$$
$$|10\rangle \rightarrow -|10\rangle$$
$$|11\rangle \rightarrow \quad |11\rangle$$

In practice we can implement the oracle without the oracle qubit using a controlled-Z gate (ex. Show the circuit right marks the state $|10\rangle$ i.e. $|m = 2\rangle$).



Ex. Step through and check it is the same as above

15.3 Inversion about mean

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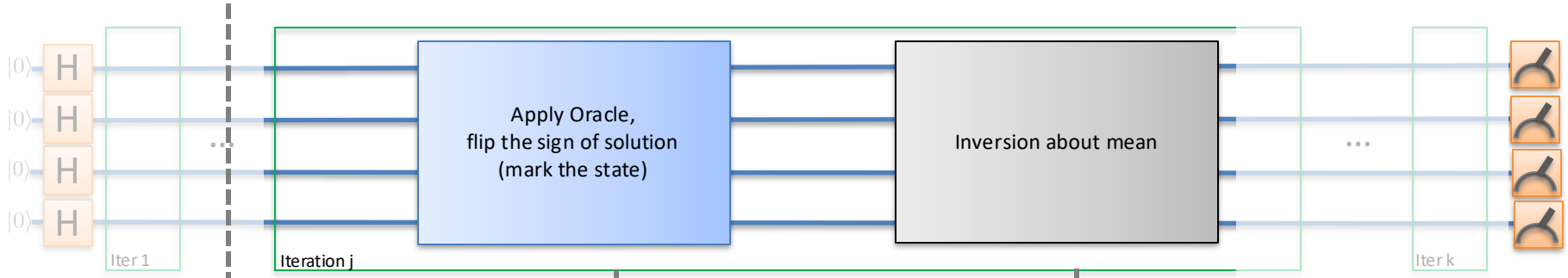
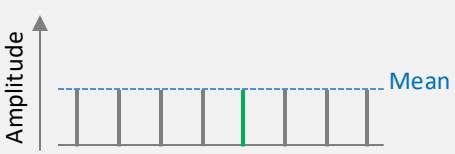
Lecture 15

Grover's algorithm: Intuition

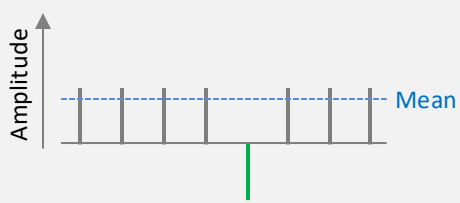


Maps every amplitude Mean$-\delta$ to Mean$+\delta$

Alice's coffee problem

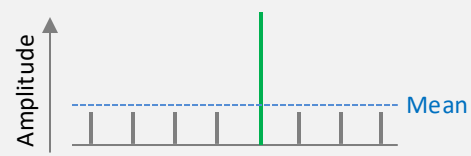


0 1 2 3 4 5 6 7



0 1 2 3 4 5 6 7

Flipping the sign of the amplitude of solution vector reduces the mean.



0 1 2 3 4 5 6 7

Inversion about the mean increases the amplitude of solution vector.

Intuition:

Grover's algorithm: Inversion around mean

- Maps every amplitude Mean$-\delta$ to Mean$+\delta$
- Consider a component $a_i|x_i\rangle$ of the overall state

  Let $a_i = A - \delta_i$ where A is the mean of amplitudes.

  Alternatively, $\delta_i = A - a_i$.

  Therefore, $A + \delta_i = A + (A - a_i) = 2A - a_i$

- Equivalent transformation:

$$\sum_{i=0}^{N-1} a_i|x_i\rangle \rightarrow \sum_{i=0}^{N-1}(2A - a_i)|x_i\rangle$$

Aside - Grover's algorithm: Inversion around mean

- Transformation is linear.

$$\sum_{i=0}^{N-1} a_i|x_i\rangle \rightarrow \sum_{i=0}^{N-1}(2A - a_i)|x_i\rangle \quad \Rightarrow \quad \begin{bmatrix} \frac{2}{N}-1 & \frac{2}{N} & \cdots & \frac{2}{N} \\ \frac{2}{N} & \frac{2}{N}-1 & \cdots & \frac{2}{N} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{2}{N} & \frac{2}{N} & \cdots & \frac{2}{N}-1 \end{bmatrix}$$

We will call this matrix D

- A quick calculation convinces us of this:

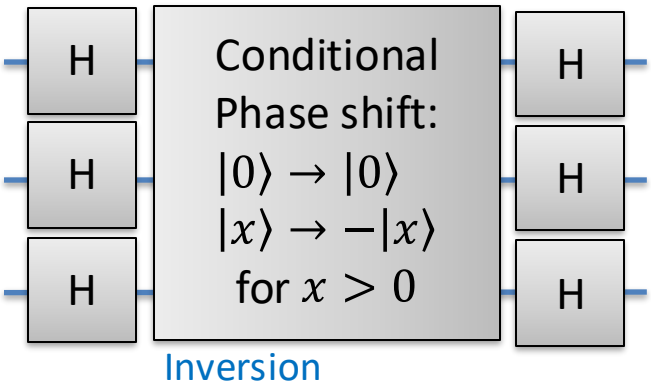
$$\begin{bmatrix} \frac{2}{N}-1 & \frac{2}{N} & \cdots & \frac{2}{N} \\ \frac{2}{N} & \frac{2}{N}-1 & \cdots & \frac{2}{N} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{2}{N} & \frac{2}{N} & \cdots & \frac{2}{N}-1 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{N-1} \end{bmatrix} = \begin{bmatrix} a`_0 \\ a`_1 \\ \vdots \\ a`_{N-1} \end{bmatrix}$$

$$\begin{aligned} a`_0 &= a_0\left(\frac{2}{N}-1\right) + a_1\left(\frac{2}{N}\right) + \cdots + a_{N-1}\left(\frac{2}{N}\right) \\ &= a_0\left(\frac{2}{N}\right) + a_1\left(\frac{2}{N}\right) + \cdots + a_{N-1}\left(\frac{2}{N}\right) - a_0 \\ &= 2\left(\frac{a_0+a_1+\cdots+a_{N-1}}{N}\right) - a_0 \\ &= 2A - a_0 \end{aligned}$$

This is not examinable (interest only)

Aside - Grover's algorithm: Inversion around mean

- This transformation can be implemented in $O(n) = O(log_2(N))$ gates.



- The matrix D can be implemented as $-H^{\otimes n}\, S_0\, H^{\otimes n}$ where S_0 is the matrix below.

$$\begin{bmatrix} -1 & 0 & 0 & 0 & \dots \\ 0 & 1 & 0 & 0 & \dots \\ 0 & 0 & 1 & 0 & \dots \\ 0 & 0 & 0 & 1 & \dots \\ \dots & \dots & \dots & \dots & \dots \end{bmatrix}$$

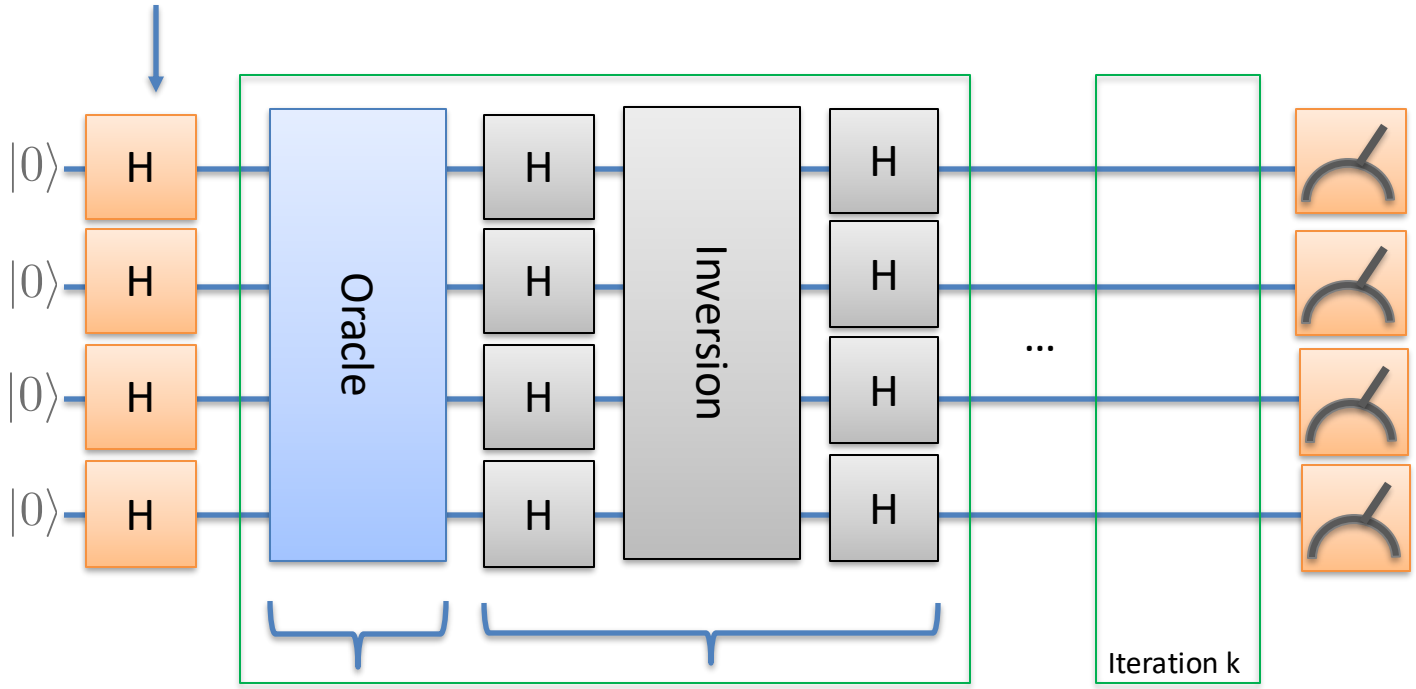
Phase inversion of $|0\rangle$ vector.

- For the curious (the derivation below is not examinable).
  - Operator corresponding to the phase shift is $2|0\rangle\langle 0| - I$
  - $H^{\otimes n}\,(2|0\rangle\langle 0| - I)\,H^{\otimes n}$ = $2|\psi\rangle\langle\psi| - I$ where $|\psi\rangle$ is the equal superposition of states.
  - $2|\psi\rangle\langle\psi| - I$ applied to general state $\sum_{i=0}^{N-1} a_i|x_i\rangle$ yields $\sum_{i=0}^{N-1}(2A - a_i)|x_i\rangle$

Grover's algorithm: Inversion around mean

Equal superposition

Solution has higher probability



The oracle

"Inversion about the mean"

One iteration of Grover's algorithm

15.4 Example Quantum Search

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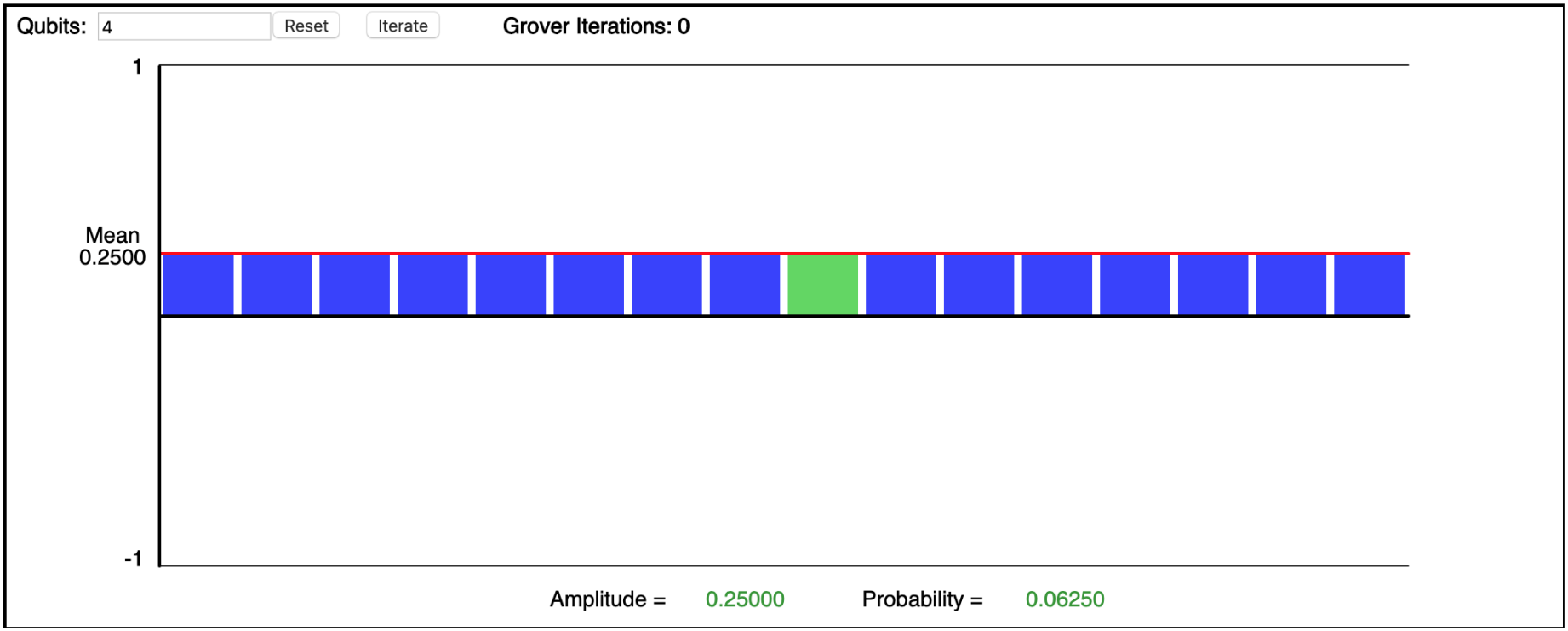
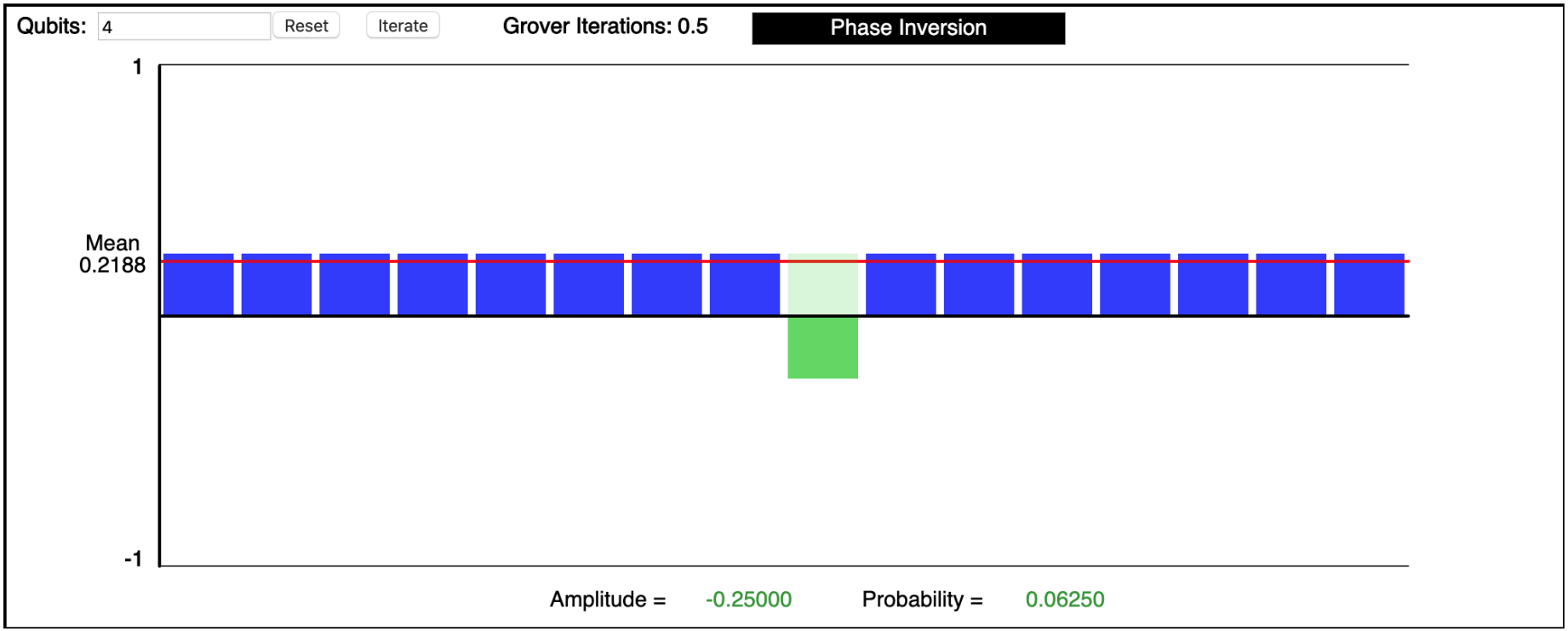
Lecture 15

Grover search with one marked state



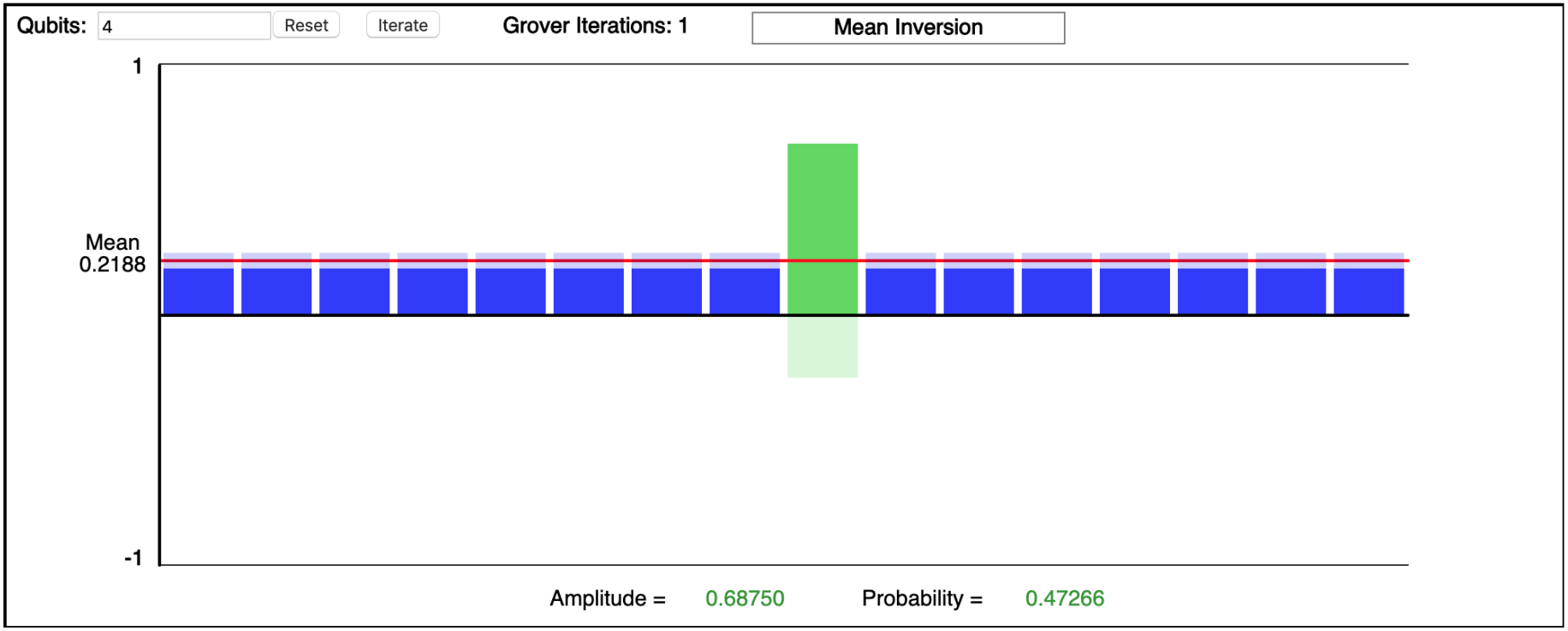
https://codepen.io/samtonetto/full/BVOGmW

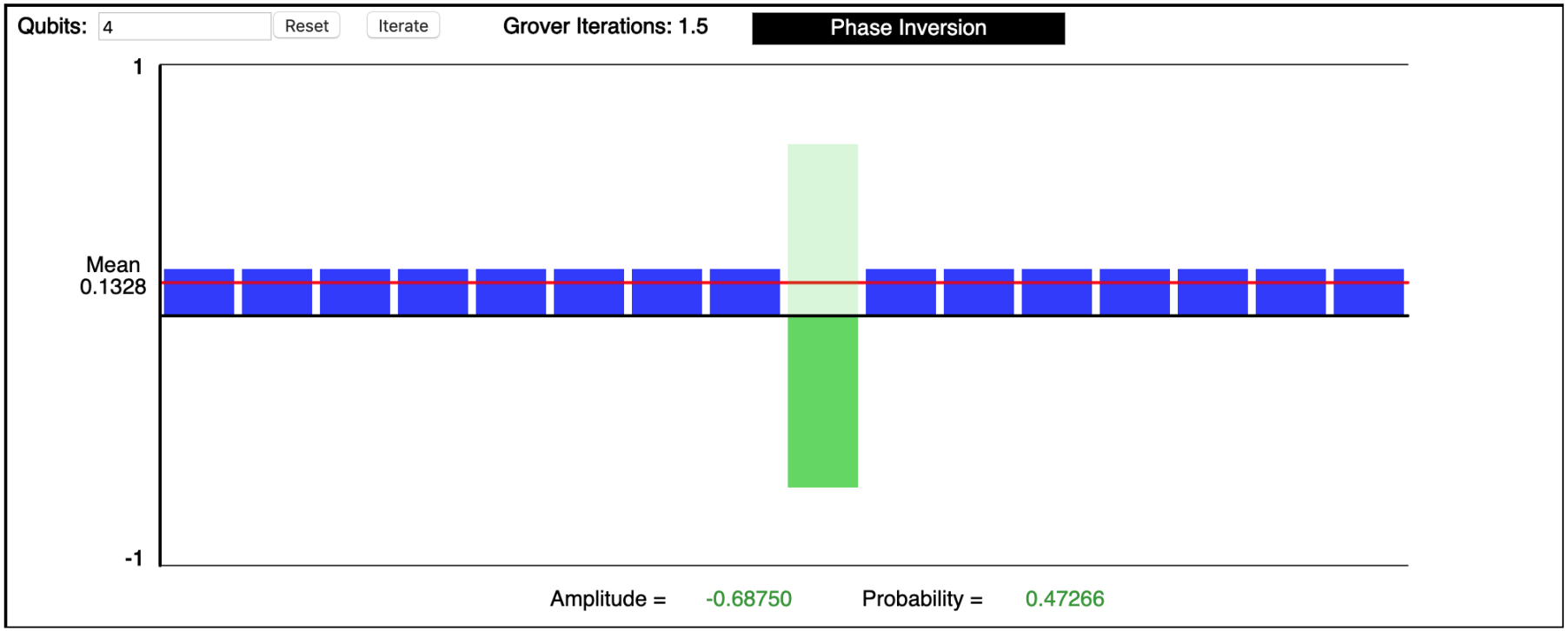
Invert the marked state

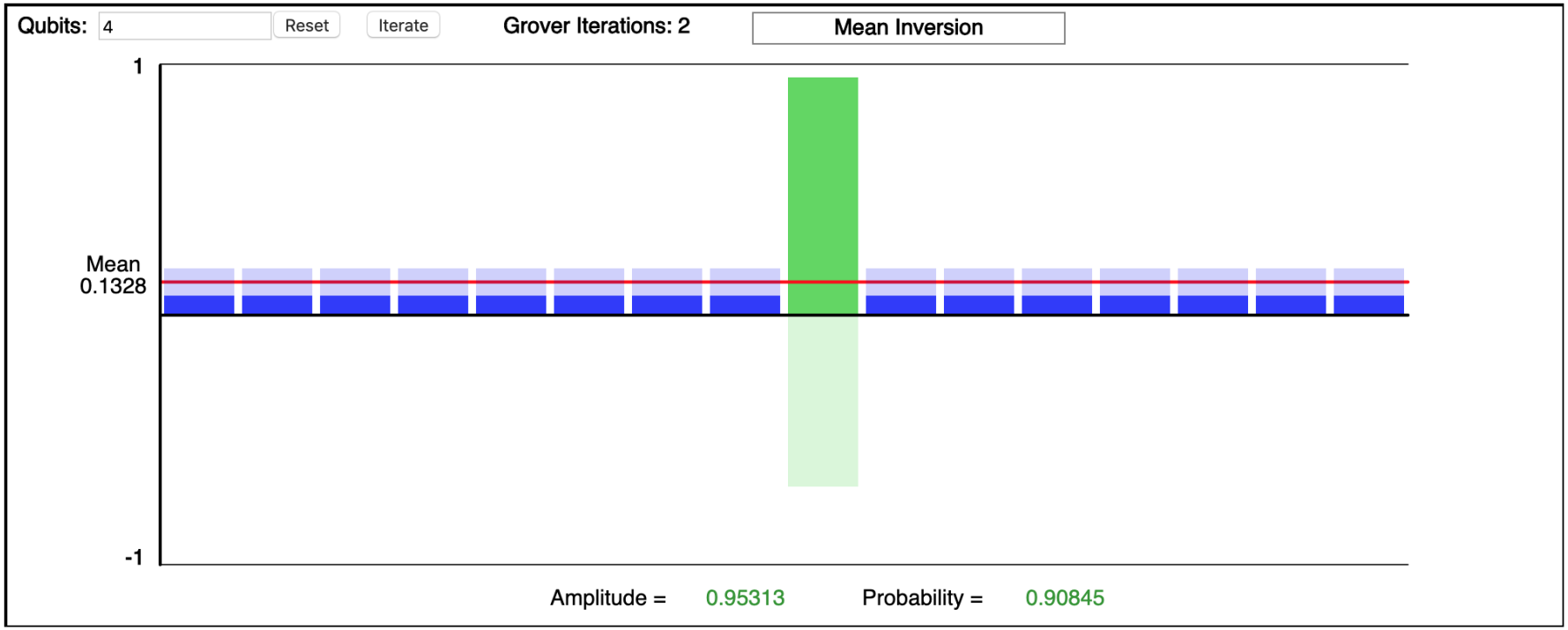


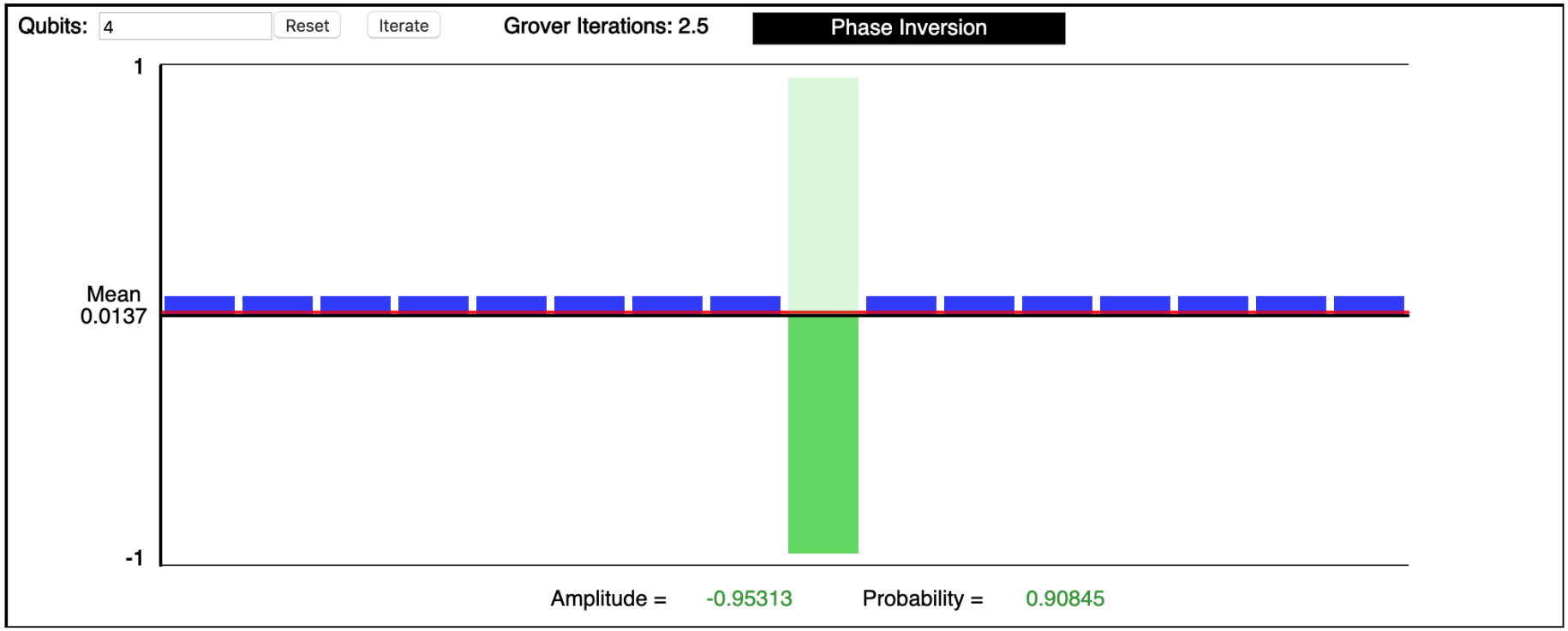
Note: The mean is reduced. Next step is inversion about the mean.

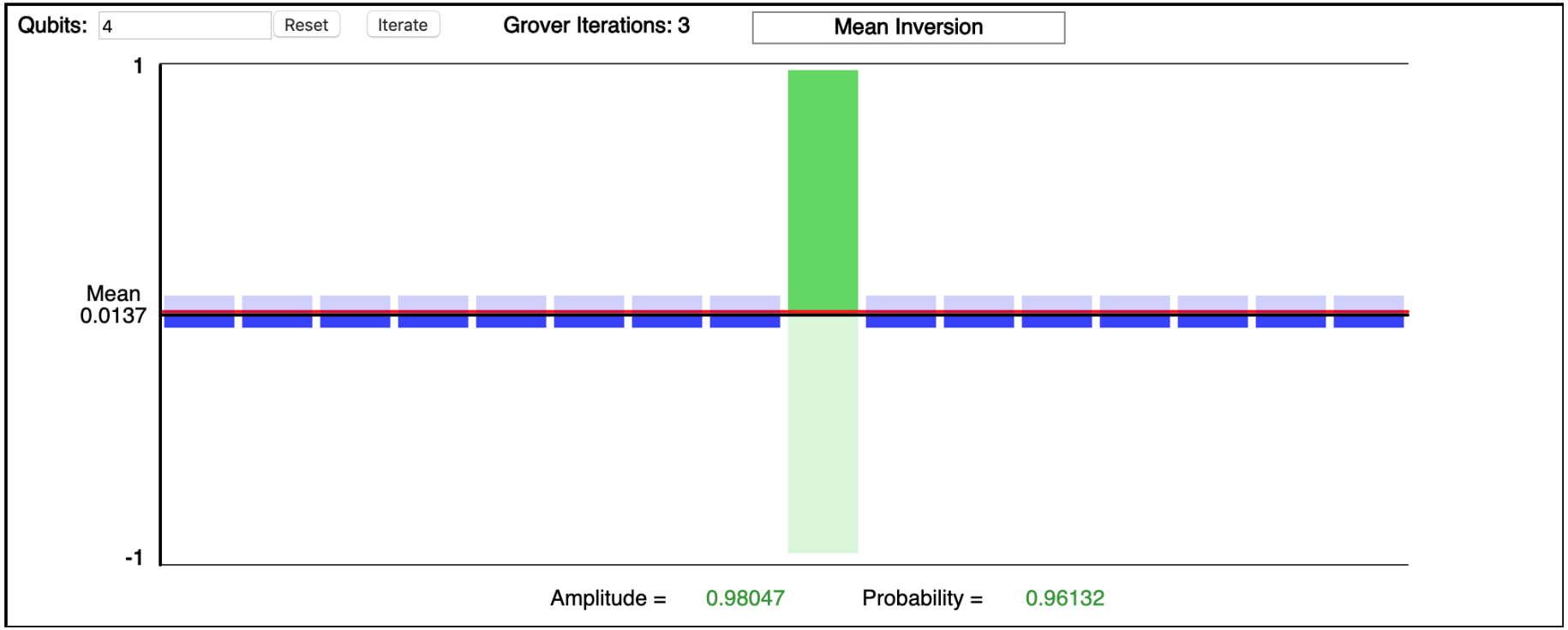
Inversion about the mean




Qubits: 4
Reset
Iterate
Grover Iterations: 1.5
Phase Inversion
1
Mean
0.1328
-1
Amplitude = -0.68750
Probability = 0.47266


Qubits: 4
Reset
Iterate
Grover Iterations: 2
Mean Inversion
1
Mean
0.1328
-1
Amplitude = 0.95313
Probability = 0.90845


Qubits: 4
Reset
Iterate
Grover Iterations: 2.5
Phase Inversion
1
Mean
0.0137
-1
Amplitude = -0.95313
Probability = 0.90845


Qubits: 4
Reset
Iterate
Grover Iterations: 3
Mean Inversion
1
Mean
0.0137
-1
Amplitude = 0.98047
Probability = 0.96132

Week 8

Lecture 15

15.1 Grover's algorithm: Intuition

15.2 The Oracle

15.3 Inversion about the mean

15.4 Example Quantum Search

Lecture 16

16.1 Grover's algorithm Examples

16.2 Probability of success

16.3 Geometric interpretation

Practice class 8

Grover's algorithm using QUI

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